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JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9648/02

Paper 2 Core Paper **26 August 2016**

Additional Materials: Answer Paper **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the question paper.

Section B
Answer any **one** question on the answer paper provided.
Circle the question number of the question attempted.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
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6	
7	
8	
Section B	
9 / 10	
Total	

This document consists of **11** printed pages and **1** blank page.

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Section A

Answer **all** the questions in this section.

1 Cathepsin is a class of protease that is commonly found in the lysosomes of all animals.

Different subtypes of cathepsin had been identified, each with its specific substrate and optimum pH. Fig 1.1 shows the relative activity of three subtypes of cathepsin at different pH.

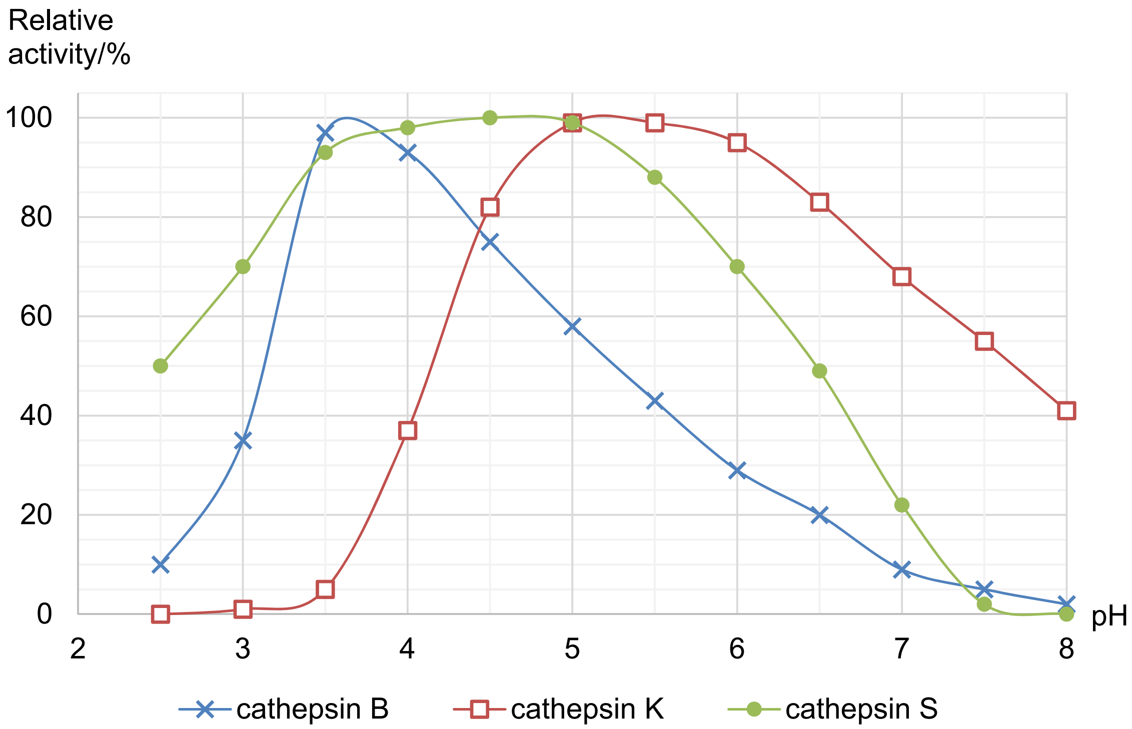


Fig. 1.1

(a) With reference to Fig 1.1,

(i) explain why cathepsin K is not active at pH 3; [3]

1. **Low pH <u>alters the ionic charge</u> of the acidic and basic R groups of amino acids at the active site of cathepsin K, thus <u>ionic bonds/hydrogen bonds</u> that help to maintain the specific shape of the active site are <u>disrupted</u>;;**
2. **leading to <u>loss of 3-D conformation of active site</u> of cathepsin K and loss of 3-D conformation of cathepsin K, cathepsin K are <u>denatured</u>;;**
3. **<u>substrate can no longer bind to active site</u> of cathepsin K to form enzyme-substrate complex, decreasing the rate of formation of E-S complexes and hence decreasing the rate of reaction/ rate of formation of products;;**

(ii) state the optimum pH of both cathepsin B and cathepsin S; [1]

1. **Optimum pH of cathepsin B is pH 3.6 (±0.1) and the optimum pH of cathepsin S is pH 4.75 (±0.25);;**

(iii) explain the difference in optimum pH for cathepsin B and S using your knowledge of protein structure. [2]

1. **Each protein contains a <u>precise number, type and sequence of amino acids</u> that are held together by peptide bonds in a linear strand of polypeptide chain;;**
2. **The number and types of <u>bonds</u> that maintain the <u>secondary</u> and <u>tertiary structures</u> of these two enzymes are <u>different</u> and these bonds remain intact at different pH;;**

Different subtypes of cathepsins are expressed in different cell types. There is sufficient evidence that overexpression of cathepsin leads to diseases. For example, high level of cathepsin K has been implicated with development of osteoporosis.

For the purpose of treatment, inhibitor molecules are designed to specifically target cathepsin. These inhibitors are categorised into two broad groups: competitive inhibitors or non-competitive inhibitors.

(b) Fig. 1.2 shows effect of substrate concentration on the activity of cathepsin in the absence of inhibitor.

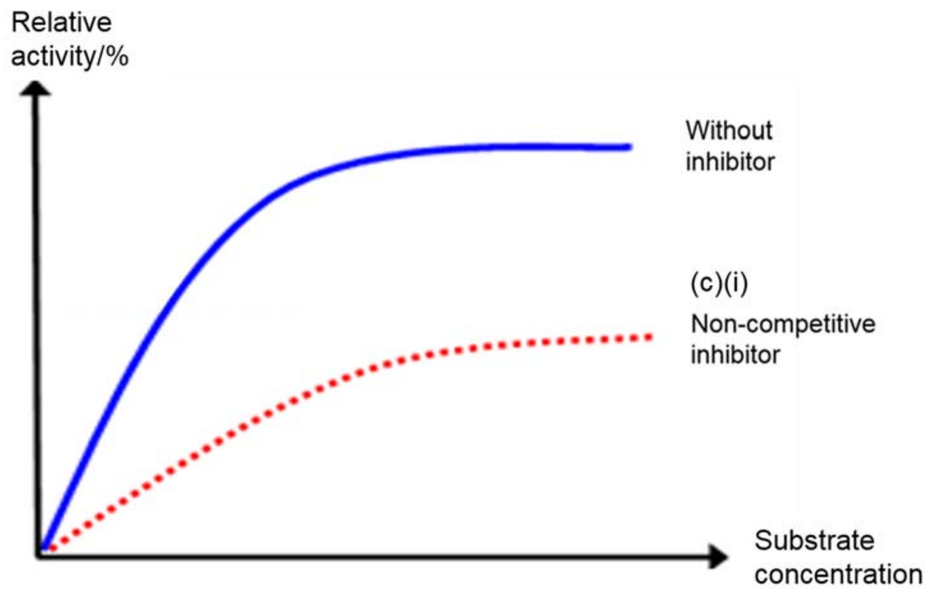


Fig. 1.2

(i) On Fig. 1.2, draw the curve you would expect in the presence of a non-competitive inhibitor. [1]

(ii) Explain with reasons the shape of the curve you have drawn. [3]

1. **Non-competitive inhibitor has no structural similarity to the substrate and <u>binds</u> to the cathepsin <u>at site other than the active site</u>;;**
2. **Binding of the non-competitive inhibitor cause cathepsin's 3D conformation to change such that its <u>active site's conformation is altered</u> and substrate can no longer bind to the active site;;**
3. **Formation of such enzyme-inhibitor complexes prevents the formation of E-S complexes and formation of products and decreasing the rate of reaction;;**
4. **An <u>increase in substrate concentration will not reverse the inhibition</u>/reduce the effect of inhibition, even at very high substrate concentration, the maximum rate of reaction in the presence of non-competitive inhibitor is lower than that of reaction in the absence of non-competitive inhibitor;;**

[Total: 10]

2 Fig. 2.1 shows the structure of the amino acids, glycine and alanine.

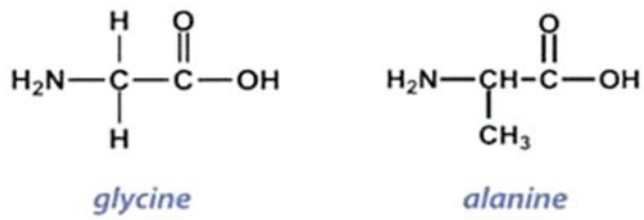
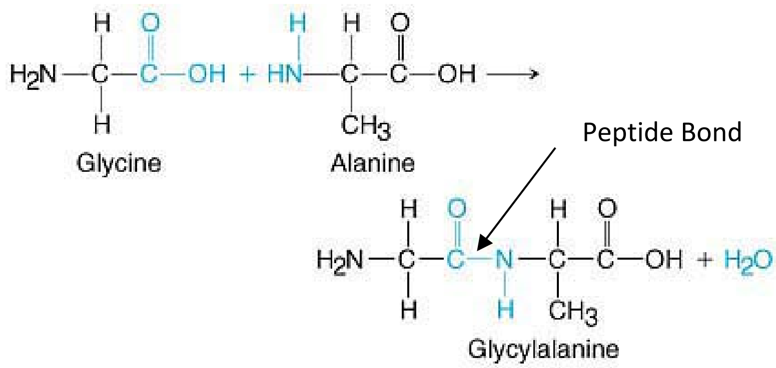


Fig. 2.1

(a) Draw in the space below, the reaction that forms a dipeptide, glycylalanine, containing the above molecules. Label the reactants and products of the reaction clearly. [2]



1. **Dipeptide with peptide bond labelled (OH and H in trans position);;**
2. **Water molecule present (correctly placed for a condensation reaction);;**
   (only 1 mark if the reactants, glycine and alanine not in the drawing)

Glycine and alanine are two of the many amino acids in the structure of haemoglobin, a red pigment that functions in the transport of oxygen in red blood cells. Haemoglobin is a soluble and stable molecule in solution.

(b) Explain how the properties and the location of the amino acid residues in a haemoglobin molecule help maintain the solubility and stability of haemoglobin in solution. [3]

1. **<u>Hydrophilic R groups</u> of amino acid residues <u>point outwards</u> and their <u>hydrophobic R groups</u> point into the centre of the molecule;;**
2. **Outward-pointing hydrophilic R groups on the surface of the haemoglobin molecule interact with the <u>polar water molecules</u>;;**
3. **Hydrophobic R groups inside the haemoglobin molecule interact in a <u>non-polar</u> environment, holding the molecule in its <u>correct three-dimensional conformation / shape</u>;;**

(c) Haemoglobin consists of 2 α-globin chains (each of 141 amino acids) and 2 β-globin chains (each of 146 amino acids). The *HBA* gene that codes for the α-globin protein is found on chromosome 16 of humans and the *HBB* gene that codes for the β-globin protein is found on chromosome 11. The mRNA molecules involved in the synthesis of the globin proteins are shorter compared to the original *HBA* and *HBB* genes.

(i) Explain why the mRNA molecules are shorter compared to the original *HBA* and *HBB* genes. [1]

1. **RNA splicing where the introns were excised and the exons joined together;;**

(ii) State other changes that can occur to mRNA molecules after they are synthesised. [1]

1. **5' capping (7-methylguanosine cap) and 3' polyadenylation (poly A tail);;**

In some forms of β-thalassaemia, an inherited blood disorder characterised by the abnormal formation of haemoglobin, the translation of the β-globin gene is affected. Truncated forms of the β-globin polypeptides were found in thalassaemia patients.

(d) Using your biological knowledge,

(i) describe the termination stage during normal translation of the β-globin mRNA; [2]

1. **Stop codon (e.g. UAA, UAG and UGA) on mRNA encountered by the ribosome, ribosome stops translocating;;**
2. **Protein release factor would recognise and bind to the stop codon on the mRNA;;**
3. **Addition of water molecule causes hydrolysis/cleavage of the ester linkage of the fully-translated polypeptide from the tRNA;;**
4. **releases polypeptide / β-globin chain from ribosome;;**
   Point 1 needed for full credit [max 2]

(ii) suggest and explain the type of mutation that caused the presence of the truncated forms of β-globin polypeptides found in the thalassaemia patients. [1]

1. **Nonsense mutation that changed a codon coding for an amino acid into a stop codon, leading to the premature termination of translation and resulting in truncated / shortened polypeptides;;**

[Total: 10]

3 Fig. 3.1 shows the main structural features of a mature (infective) human immunodeficiency virus (HIV).

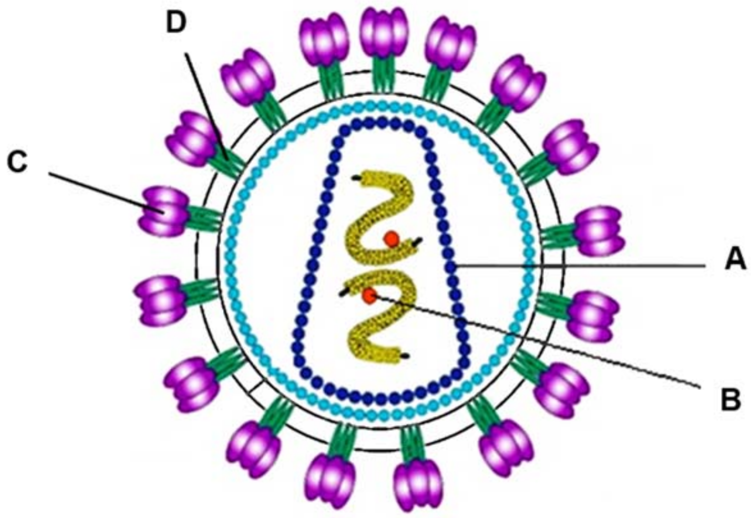


Fig. 3.1

(a) Identify the structure **A** and **B**. [2]

1. **A – capsomere/capsid protein;;**
2. **B – reverse transcriptase/integrase;;**

(b) Describe the role of **C** and **D** during the entry of HIV into T cells. [3]

1. **<u>gp120</u> on HIV envelope <u>recognizes and binds to a CD4 molecule</u> on the plasma membrane of the T cell;;**
2. **The interaction between gp120 and CD4 induces a <u>conformation change in gp120</u>, enabling the <u>binding</u> of gp120 to the host cell's <u>chemokine receptor</u>, <u>CCR5 or CXCR4</u>;;**
3. **The interaction between gp120 and chemokine receptor brings about a <u>conformation change in gp41</u> on the HIV virion, allowing the <u>fusion of the HIV envelope and plasma membrane of T cells</u>;;**

(c) As the global incidence of HIV exceeds 2 million new infections annually, effective interventions to decrease HIV transmission are needed. Truvada is a combination tablet contains two different types of drugs: tenofovir and emtricitabine. Both drugs inhibit reverse transcriptase using different mechanisms.

Studies have demonstrated that daily consumption of Truvada tablets protects healthy individuals against HIV infection by reducing the number of virus copies in the blood.

(i) Using the information above and your knowledge of HIV life cycle, explain how Truvada reduces the number of virus copies in the blood. [2]

1. **Truvada targets the <u>replication</u> stage of HIV by inhibiting the synthesis of double stranded DNA from viral RNA by reverse transcriptase;;**
2. **There is no incorporation of double stranded DNA into the host cell's DNA and no provirus formed;;**
3. **No synthesis of viral protein and enzymes and thus, no assembly and release of new HIV virus;;**

(ii) Suggest why two inhibitors with different mechanisms are combined into Truvada. [2]

1. **(Point) mutations in the genes coding for reverse transcriptase altering its active site/allosteric site/tertiary structure resulting in the inhibitors cannot bind to these sites/inhibit the activity of viral enzyme/HIV being resistant to one of the inhibitors;;**
2. **(As the two mutations of same gene are unlikely to occur at the same time) Taking a mixture of two drugs prevents/reduces the chance of drug-resistant HIV strain from developing;;**

Fig. 3.2 shows the structure of a Zika virus (ZikV). Most ZikV infection shows no symptom while other infections result in Zika fever which is characterised by mild symptoms such as fever, conjunctivitis (eye infection) and joint pain.

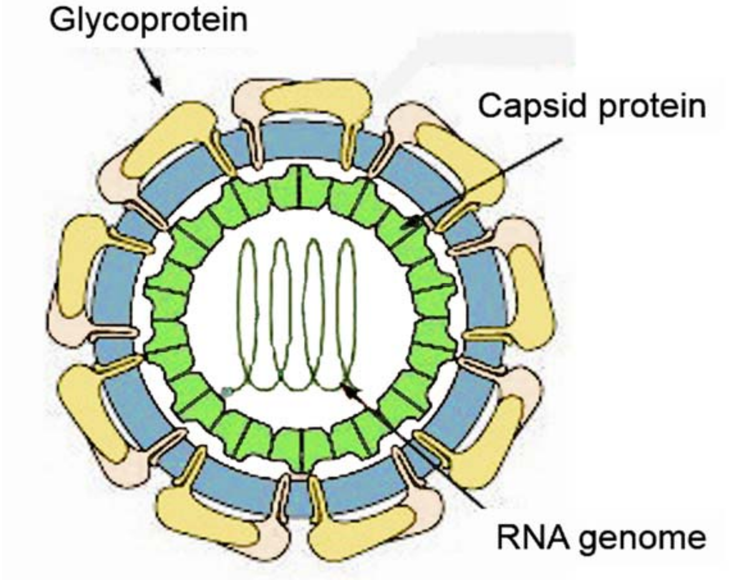


Fig. 3.2

(d) With reference to Fig. 3.1 and Fig. 3.2, state two structural differences between HIV and ZikV. [2]

1. **HIV has two RNA segments while ZikV has one RNA segments;;**
2. **Presence of enzymes within HIV but absence of enzymes within ZikV;;**

(e) During the ongoing ZikV outbreak, the incidence of Guillain-Barré syndrome had increased by four times. Guillain-Barré syndrome is a disorder in which the immune system attacks peripheral nervous system leading to progressive muscle weakness.

Using the information above, suggest an explanation on how ZikV infection could have resulted in higher incidence of Guillain-Barré syndrome. [1]

1. **ZikV infects the nerve cells/neurons and triggers the activation of immune system to destroy the viral-infected nerve cells;;**

[Total: 12]

4 The Philadelphia chromosome is an abnormality in chromosome 22 found in chronic myeloid leukemia. This chromosome is defective and unusually short because of reciprocal translocation between chromosome 9 and chromosome 22.

Fig 4.1 shows the chromosome translocation.

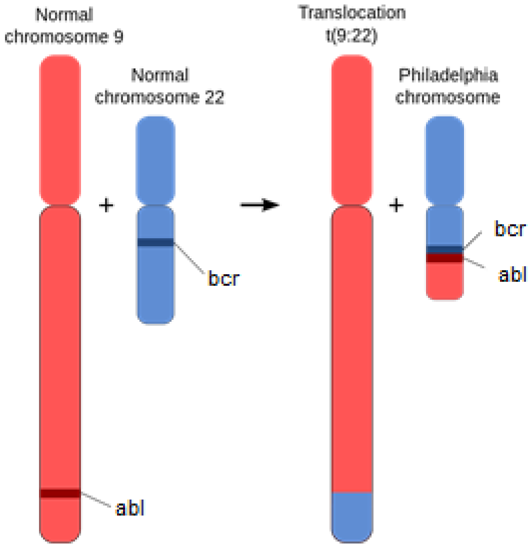


Fig. 4.1

abl is a proto-oncogene found on chromosome 9. Translocation results in an oncogene, *bcr-abl* gene fusion that can be found on the shorter derivative chromosome 22.

Majority of patients with chronic myeloid leukemia have breakpoints in part of *abl* gene and in part of *bcr* gene resulting in *bcr-abl* gene fusion. The bcr-abl transcript is translated into a tyrosine kinase that is constitutively active.

(a) State what is meant by an oncogene. [2]

1. <u>Mutated form of proto-oncogene</u>;;
2. Oncogene expression leads to an increase in <u>amount/ intrinsic activity of the proto-oncogene protein</u>, stimulating excessive cell division;;

(b) Explain how the *abl* proto-oncogene becomes an oncogene in chronic myeloid leukemia cells. [3]

1. *abl* proto-oncogene is translocated to chromosome 22 and fuses/joins with *bcr* gene;;
2. Change in nucleotide sequences;;
3. Results in a <u>frameshift mutation</u> where all codons downstream of the mutation will be changed;;
4. As a consequence, *bcr-abl* gene product/ tyrosine kinase would have a <u>different sequence and number of amino acids</u> which is <u>constitutively active</u>;;

Reject: under the control of active promoter/ amplification of abl-bcr gene

The *H-ras* oncogene codes for the ras protein.

Fig.4.2 is part of the base sequence of a proto-oncogene and of the *H-ras* oncogene, which arises from the proto-oncogene by mutation.

Fig. 4.2 also shows part of the primary structure of their two encoded proteins.

proto-oncogene

```
ATG ACG GAA TAT AAG CTG GTG GTG GTG GGC GCC GGC GGT GTG GGC
met thr glu tyr lys leu val val val gly ala gly gly val gly
```

encoded protein

oncogene

```
ATG ACG GAA TAT AAG CTG GTG GTG GTG GGC GCC GTC GGT GTG GGC
met thr glu tyr lys leu val val val gly ala val gly val gly
```

encoded protein

Fig. 4.2

(c) Predict how ras protein differs from the normal protein encoded for by proto-oncogene. [1]

1. **More active;; OR**
2. **More resistant to degradation;;**

(d) With reference to Fig. 4.1 and Fig. 4.2, describe how this mutation of proto-oncogene to oncogene differs from the mutation that involves the *bcr-abl* oncogene. [2]

1. **Mutation involving *abl* gene is a form of <u>chromosomal mutation</u> (translocation) where a <u>portion of chromosome 9 is transferred to chromosome 22</u>;;**
2. **mutation involving *ras* gene is a form of <u>gene mutation</u> where <u>guanine is substituted for by thymine</u>;;**

[Total: 8]

5 Onion bulbs may be red, white or yellow. These colour differences are under the control of two pairs of genes.

A farmer crossed a plant with white onions and a plant with red onions. All F_1 offspring were plants with red onions. The offspring were selfed and the results of the cross were:

Plants with red onions	182
Plants with yellow onions	65
Plants with white onions	82

(a) Using a genetic diagram, illustrate the selfing of the F_1 generation to give the F_2 generation. [4]

Use the following symbols:
A represents allele coding for enzyme which deposits pigment
a represents allele coding for non-functional enzyme for pigment deposition
B represents allele for red pigment
b represents allele for yellow pigment

F_1 phenotype	**plants with red onions**	**x**	**plants with red onions**
F_1 genotype	**AaBb**	**x**	**AaBb ;;**
Gametes ;;	**AB Ab aB ab**		**AB Ab aB ab**

Fertilization

	AB	**Ab**	**aB**	**ab**
AB	**AABB**	**AABb**	**AaBB**	**AaBb**
Ab	**AABb**	**AAbb**	**AaBb**	**Aabb**
aB	**AaBB**	**AaBb**	**aaBB**	**aaBb**
ab	**AaBb**	**Aabb**	**aaBb**	**aabb**

Genotypes	**A_B_**	**A_bb**	**aa _ _ ;;**
Phenotypes	**Red onions**	**Yellow onions**	**White onions**
Phenotypic ratio	**9:**	**3:**	**4 ;;**

(b) A chi-squared test was performed on the results of the cross.

(i) Calculate the χ^2 value by filling in the Table 5.1. [2]

Table 5.1

	Observed (O)	Expected (E)	$\frac{(O-E)^2}{E}$	$\sum \frac{(O-E)^2}{E}$
Plants with red onions	182	**185**	**0.0486**	χ^2 value = **0.194**
Plants with yellow onions	65	**62**	**0.145**	
Plants with white onions	82	**82**	**0**	

Award 1 mark for each correct column for expected and $\frac{(O-E)^2}{E}$

(ii) Suggest reasons why the actual numbers observed do not match the expected ratio exactly. [2]

1. **Due to random fertilisation/chance, the numbers were not exactly as predicted;;**
2. **The sample size would need to be larger to get closer to the expected ratio;;**

(iii) Using Table 5.2, explain if the observed numbers conformed to the expected. [2]

Table 5.2

degrees of freedom	probability P				
	0.50	0.10	0.05	0.01	0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.61	5.99	9.21	13.82
3	2.37	6.25	7.82	11.35	16.27
4	3.36	7.78	9.49	13.28	18.47

1. **The calculated χ^2 value is 0.194, less than the critical χ^2 value of 5.99 at p= 0.05, Value of p is more than 0.50/ more than p = 0.05;;**
2. **Do not reject the null hypothesis, there is no significant difference between the observed and the expected ratio;;**
   Observed numbers conformed to the expected.

[Total: 10]

6 Telomeres are DNA nucleotide sequences found at both ends of chromosome. The telomeres in somatic cells will get slightly shorter with each round of cell division until a critical short length is reached. Critically short telomeres trigger activation of p53 protein which is required for the expression of a pro-apoptosis gene, P53 Up-regulated Modulator of Apoptosis (*puma*).

Fig. 6.1 shows part of the process, transcription of *puma* taking place.

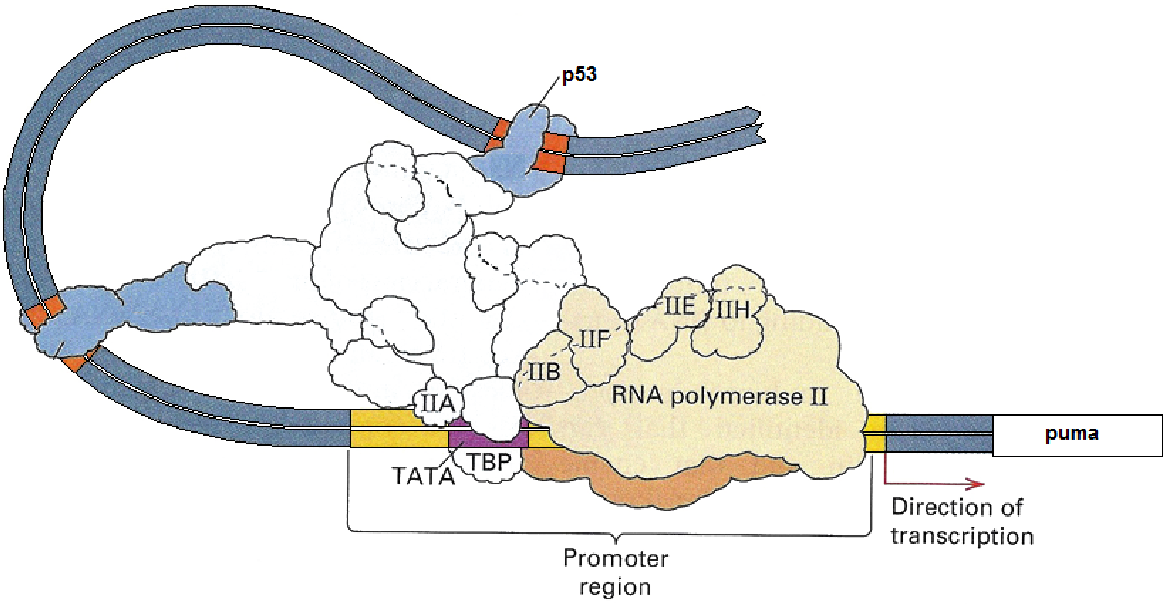


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) suggest the role of p53; [1]

1. acts as an activator/specific transcription factor;;

(ii) describe the initiation of transcription of *puma*. [4]

1. **<u>TATA-binding protein recognizes and binds to TATA box</u> within the promoter region;;**
2. **General transcription factors bind to the TATA-binding protein, recruiting RNA polymerase to assemble, forming transcription initiation complex;;**
3. **<u>p53</u> /activator recognizes and binds to <u>enhancer</u> causing <u>DNA bending</u>;;**
4. **bringing p53 in contact with the transcription initiation complex, initiating transcription;;**

(b) Modifications of DNA or histones facilitate the expression of *puma*.

Outline how two modifications of DNA or histones can lead to expression of *puma*. [3]

1. <u>Acetylation of histones</u> by histone acetyltransferase removes positive charges on histones resulting in the loosening of histone complex from DNA;;
2. <u>Demethylation of DNA</u> by DNA demethylase, resulting in chromatin becoming less tightly coiled;;
3. <u>Chromatin remodeling complex</u> alter structure of nucleosomes such that chromatin is less tightly coiled;;
Max 2 for pt 1-3

4. allows RNA polymerase and transcription factors to access the promoter to initiate transcription of puma gene;;

(c) Explain the significance of activating p53 in the presence of critically short telomeres. [2]

1. With further rounds of cell division/DNA replication, telomeres will shorten further, eroding important genes causing mutations;;
2. Activating p53 triggers apoptosis for cells with critically short telomeres can <u>prevent cells from accumulating mutations</u>;;

[Total: 10]

7 Fig. 7.1 shows part of a wheat leaf cell under high power magnification of an electron microscope.

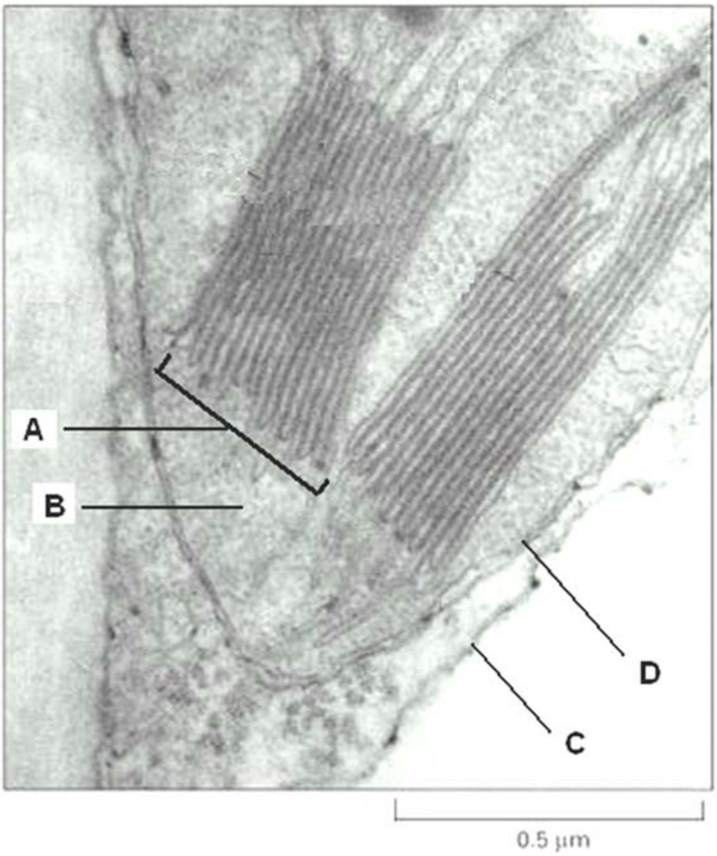


Fig. 7.1

(a) Identify the structures labelled **A** to **D**. [2]

A - Granum (reject grana);
B - Stroma;
C - Cell wall / cell membrane;
D - Chloroplast envelope / membrane;

In wheat plants, the light-dependent reactions are carried out by photosynthetic pigments which fall into two categories: primary pigments and accessory pigments.

(b) Using your biological knowledge,

(i) explain the significance of photosynthetic membranes in chloroplasts to photosynthetic pigments; [1]

1. **Hold the photosynthetic pigments in a suitable position for absorbing the maximum amount of light energy;;**
2. **Provides a large surface area which holds the photosynthetic pigments;;**
   [max 1]

(ii) outline the role played by accessory pigments in the light-dependent reactions. [2]

1. **Strongly absorb wavelengths of light which cannot be captured by chlorophyll;;**
2. **Transfer the energy from these wavelengths of light to special chlorophyll a / reaction centre;;**

(c) Wheat is usually grown in the open fields for agricultural purposes. During one of the growing seasons, the skies were unusually dark and gloomy. It was observed that the wheat was not able to develop fully.

Suggest a possible explanation for the above observation. [3]

1. **Light is acting as the <u>limiting factor</u> and is light energy is needed during the <u>light dependent reactions</u>;;**
2. **to excite and boost electrons in the reaction centre of the photosystems to a higher energy level;;**

3. **With <u>less light, fewer electrons</u> would be <u>excited</u>, and <u>less photophosphorylation</u> can occur leading to <u>less</u> <u>ATP and NADPH</u> formed;;**
4. **therefore light independent stage would be less active / <u>Calvin cycle</u> will not occur, thus no formation of GP and thus no formation of TP, hence no glucose and hence cellulose formation → no growth;;**

NADP is an important molecule in the light-dependent and light-independent reactions of photosynthesis.

(d) Outline the role of NADP in wheat leaf cells. [2]

1. **Acts as a <u>coenzyme</u> (for dehydrogenase) / acts as hydrogen (atom) / hydrogen ion and electron <u>carrier</u>/acceptor;;**
2. **NADP is <u>reduced to NADPH</u> and <u>NADPH carries electrons and protons/hydrogen atom</u> (from light dependent reactions) to <u>light independent</u> reactions/Calvin cycle / provides reducing power for the light independent reactions / Calvin cycle;;**
3. **for the <u>reduction of GP to TP</u>;;**
4. **leading to the <u>regeneration of NADP</u> (for subsequent light dependent reactions);;**
[max 2]

[Total: 10]

8 Anole lizards are found throughout the Carribbean and the surrounding mainland. There are many species of these lizards. Each species is found only on one island or a small group of islands, apart from *Anolis carolinensis* which is found in mainland Florida, USA.

Fig. 8.1 shows the distribution of four species of anole lizards.

Fig. 8.1

An investigation was carried out into the relationships between these four lizard species, using DNA analysis. The base sequences of a region of mitochondrial DNA (mtDNA) from the four species were compared. The results are shown in Table 8.1.

The smaller the number in the boxes, the smaller the percentage differences between the base sequences of the two species compared.

	A. brunneus			
A. brunneus		*A. smaragdinus*		
A. smaragdinus	12.1		*A. carolinensis*	
A. carolinensis	16.7	15.0		*A. porcatus*
A. porcatus	11.3	8.9	13.2	

Table 8.1

(a) With reference to Table 8.1, state the species to which *A. brunneus* appears to be most closely related. [1]

1. ***A. porcatus*;; (to underline)**

(b) Researchers put forward the hypothesis that the three species, *A. brunneus, A smaragdinus and A. carolinensis*, have originated from three **separate** events in which a few individuals of *A. porcatus* spread directly from Cuba to three different places.

Explain how the results in Table 8.1 support the researchers' hypothesis. [3]

1. ***A. brunneus, A smaragdinus* and *A. carolinensis* have smaller percentage differences with *A. porcatus* (than with others);;**
2. **Quote figures e.g. 11.3%, 8.9%, 13.2%;;**
3. **More closely related to *A. porcatus* (than to each other);;**
4. **Quote figures e.g. *A. brunneus* with *A. smaragdinus* have 12.1% differences, *A. brunneus* with *A. carolinensis* have 16.7% differences;;**
   [max 3]

(c) Explain how a population of *A. porcatus* that became isolated on an island could have evolved into a new species. [4]

1. **The subpopulations of *A. porcatus* became <u>geographically isolated by the water barrier leading to allopatric speciation</u>;;**
2. **The subpopulations <u>did not interbreed</u> (reproductive isolation) and thus <u>gene flow was disrupted</u>;;**
3. **The subpopulations were exposed to <u>different environments</u> on the different islands and were thus subjected to <u>different selection pressures</u> e.g. food availability / temperature / different predators etc;;**
4. **Since there was <u>variation within the subpopulations</u>, lizards with <u>favourable characteristics were selected for / individuals with favourable characteristics had a selective advantage</u>;;**
5. **<u>Evolutionary changes</u> occurred <u>independently</u> in each subpopulation / <u>different genetic changes</u> occurred in each subpopulation. Over <u>successive generations</u>, each sub-population became a <u>genetically distinct species</u>;;**

(d) Explain why some phylogenetic studies use mitochondrial DNA (mtDNA) from mitochondria rather than nuclear DNA. [2]

1. **No recombination or crossing over takes place in the mitochondrial genome thus the differences reflect only mutations;;**
2. **If the coding region of mtDNA (e.g. cytochrome b gene) is used, it mutates very slowly as the genes in the coding region are very crucial for survival;;**
   (but the mutations are sufficient to allow for species differentiation)
3. **If non-coding mitochondrial control region of mtDNA is used, it has a more rapid mutation rate which allows for differences in DNA sequence to accumulate and allow better differentiation between closely related species for use in phylogenetic study;;**
4. **There are multiple mitochondria in a cell and hence multiple copies of mtDNA in each cell (100 – 10,000) as compared to one copy of nuclear DNA in each cell, making it easier to obtain mtDNA;;**
5. **Mitochondrial chromosomes are circular and subjected to less degradation by exonucleases;;**
   [max 2]

[Total: 10]

.

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 **(a)** Describe how the chromosomes behave during mitotic cell cycle. [6]

<u>**Interphase**</u>

1. **Chromosome exist as the uncoiled and diffused form called chromatin;;**
2. **During S phase, semi conservative DNA replication occurs and each chromatin consist of two chromatids joined by the centromere;;**

<u>**Prophase**</u>

3. **Chromatin shorten and thicken/condense and become distinct structures called chromosomes;;**

<u>**Metaphase**</u>

4. **Spindle fibres from each pole of the cell are attached to one of the two chromatids of each chromosome at the centromere region;;**
5. **Spindle fibres pull on the centromere, arranging the chromosomes in a single row on the metaphase plate (equator of the spindle);;**

<u>**Anaphase**</u>

6. **Centromere divides and sister chromatids separate at centromere, forming daughter chromosomes;;**
7. **Shortening of microtubules occurs and spindle fibres attached to the centromeres pull the daughter chromosome to the opposite poles of the cell with centromere leading the way;;**

<u>**Telophase**</u>

8. **Daughter chromosomes reach the opposite poles;;**
9. **Chromosomes uncoil, lengthen and become indistinct to form chromatin again;;**

For full credit, description of every stage is required.

(b) Outline the role of mitosis in multicellular organisms. [8]

1. **Mitosis produces genetically identical cells with the same DNA as the parent cell;;**
   R: same no and type of chromosome
2. **These genetically identical cells can be used for growth, cell replacement or for asexual reproduction;;**

   <u>Genetic stability</u>
3. **Results in genetic stability within populations of cells derived from the same parental cell;;**

   <u>Cell replacement</u>
4. **Cell division continues to function in renewal and repair. These genetically identical cells are used to replace damaged and worn out cells e.g. skin cells that die from normal wear and tear;;**
   R: repair of cells

   <u>Growth</u>
5. **Mitosis enables a multicellular organism, such as a human, to grow and develop from one single egg;;**
6. **Whole organisms grow by increasing in cell numbers by the process of mitosis to produce genetically identical somatic cells;;**

   <u>Asexual reproduction</u>
7. **Asexual reproduction is the production of new individuals of a species by one parent organism;;**
8. **Most animals and plant species are propagated by asexual reproduction resulting in clones so that the species can colonise a particular stable, well-adapted habitat within the shortest period of time;;**

   <u>Regeneration of parts</u>
9. **Some animals are able to regenerate whole / missing parts of the body, e.g. tails of house lizard, legs in crustacean and arms in starfish;;**

(c) Describe the structure and function of centromere. [6]

   <u>Structure</u>
1. **A centromere is the specialised constricted region on the chromosome where two sister chromatids are <u>joined</u> in a replicated chromosome;;**
2. **The <u>DNA</u> in the centromeric region is often <u>heavily methylated</u>;;**
3. **and includes many <u>repetitive noncoding sequences</u> known as satellite sequences;;**
4. **Consist of specific DNA sequences to which a number of centromere-associated proteins bind, forming <u>kinetochore</u>;;**

   <u>Functions:</u>
5. **Holds sister chromatids together;;**
6. **Site (on chromosome) where the kinetochore assembles and spindle fibres/microtubules from the centrioles attaches during nuclear division;;**
7. **And pull daughter chromosomes to the opposite poles of the cell;;**
8. **Ensures proper segregation of chromosomes/divides to allow separation of sister chromatids during anaphase;;**

[Total: 20]

10 (a) Explain the principles of homeostasis. [8]

1. **Homeostasis is the maintenance of a constant internal environment;;**
2. **This is achieved by countering deviation using negative feedback, which allows organism to be independent from the external factors;;**

3. **The homeostatic control system involves the stimulus, detector, control centre/regulator, effector and response;;**
   OR
4. **E.g. pancreas as the detector and regulator in the control of blood glucose concentration, liver cells or muscle cells as the effector;;**

5. **involves self-regulation and negative feedback mechanisms to control the variables that are to be maintained;;**
6. **Self-regulation involves control mechanism that is activated by the variable that it serves to regulate;;**
7. **E.g. the secretion of hormones involved in the control of blood glucose level is triggered by changes in the blood glucose level;;**
8. **Negative feedback mechanisms where any deviation of the variable from the reference point triggers mechanisms to counteract that deviation and cause the variable to return to the reference point;;**
9. **Response is in opposite direction to the stimulus and this allows blood glucose concentration to return to normal level;;**

10. **Coordination involves either the nervous system and/or the endocrine system;;**
    [max 8]

Points 1, 3, 5, 6, 8 needed for full credit

(b) Explain the role of glucagon in regulating blood glucose concentration in humans. [6]

1. **Blood glucose concentration decreases below normal level of 90mg per 100cm^3 of blood;;**
2. **Islet of Langerhans of the pancreas detect the deviation;;**
3. **α-cells of the islet of Langerhans of the pancreas secrete glucagon;;**

Points 1-3 [max 2]

4. **Glucagon is complementary in shape to specific binding site of G-protein-linked receptor;;**
5. **recognises and binds to specific binding sites of receptors on the surface membrane of liver cells;;**
6. **The activated G-protein dissociates from the receptor and activates inactive adenylyl cyclase to active adenylyl cyclase which catalyses the conversion of ATP to cyclic AMP (cAMP);;**
7. **The cAMP then acts as a second messenger and activates a phosphorylation cascade leading to activation of the enzyme glycogen phosphorylase;;**

Points 4-7 [max 2]

8. **Glycogen phosphorylase will catalyse the increased breakdown of glycogen to glucose in the liver cells / glycogenolysis;;**

9. **Glucagon stimulates an increase in the rate of conversion of amino acids and glycerol to glucose / gluconeogenesis;;**
10. **Blood glucose concentration increases and return back to normal level of 90 mg per 100 cm^3 of blood;;**

Points 8-10 [max 2]

Effects of glucagon (Points 8 and 9) must be given for full credit.

(c) Explain how a nerve impulse is passed across a synapse. [6]

A. Pre-synaptic Events [max 2]

1. **Before the presynaptic membrane is depolarised, there are few Ca^{2+} ions inside the synaptic knob;;**
2. **The arrival of an impulse/action potential at the synaptic knob causes voltage-gated Na^+ channels and voltage-gated Ca^{2+} channels to open, resulting in influx of Na^+ ions and Ca^{2+} ions into the synaptic knob from the synaptic cleft;;**
3. **The influx of Ca^{2+} ions causes vesicles containing neurotransmitters—acetylcholine to move to the presynaptic membrane and fuse with it;;**
4. **releasing the neurotransmitter into the synaptic cleft via exocytosis;;**

Point 3 need for full credit

B. Synaptic Events [1]

5. **Acetylcholine diffuse across the (20nm) synaptic cleft, (usually in less than 0.5 ms);;**

C. Post-Synaptic Events [max 3]

6. **Acetylcholine recognises and binds to specific binding sites of receptors on the postsynaptic membrane;;**
7. **This changes the shape of the receptor protein, opening chemically-gated Na^+ channels on the postsynaptic membrane;;**
8. **Influx of Na^+ ions into the cytoplasm down concentration gradient depolarise the postsynaptic membrane;;**
9. **The membrane potential becomes more positive than resting potential, and the local potential that develops is known as an excitatory postsynaptic potential (EPSP);;**
10. **If the membrane potential of the postsynaptic membrane reaches threshold potential (-50mV), an action potential is generated in the postsynaptic neurone and travels down the axon to the next synapse;;**
11. **The synaptic cleft contains an enzyme, acetylcholinesterase, which hydrolyses each acetylcholine molecule into acetate and choline. The choline is taken back into the presynaptic neurone to combine with acetyl coA to form acetylcholine;;**

Points 6 and 8 needed for full credit.

[max 6]

[Total: 20]